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Assignment - 0 Solution

Given posts :-

Post-A :- "cats are cute and funny."

Post-B :- "Dogs are funny animals."

Post-C :- "cats and dogs rarely got along."

1. Implementing Bag-of-Words :-

- Vocabulary of the corpus / Word vector $\Rightarrow$

[cats, are, cute, and, funny, dogs, animals, rarely, got, along]

- Dimension of the vector = 10.

- For, Post-A : [1 1 1 1 1 0 0 0 0 0]
Post-B : [0 1 0 0 1 1 1 0 0 0]
Post-C : [1 0 0 1 0 1 0 1 1 1] $\Rightarrow$ count vector.

- Matrix, $A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}_{3 \times 10}$

2. Designing a simple model of classification :-

$$P(\text{cat}) = p.$$

- If a post has 10 words,

$$P(\text{no occurrence of 'cat'}) = (1-p)^{10}$$

$$P(\text{at least one occurrence of 'cat'}) = 1 - (1-p)^{10}$$

If a post has L words, (of length L)

$$P(\text{cat-type}) = 1 - (1-p)^L \quad \{P(\text{at least one occurrence of 'cat'}) = P(\text{cat-type})\}$$

- Probability vector of Post-A : [0.1 0.1 0.1 0.1 0.1 0 0 0 0 0]
" " " Post-B : [0 0.1 0 0 0.1 0.1 0.1 0 0 0]
" " " Post-C : [0.1 0 0 0.1 0 0.1 0 0.1 0.1 0.1]

For Post-A,

$$P(\text{cat}) = p = \frac{1}{5} \rightarrow \frac{\text{no. of 'cat' word in post}}{\text{Total no. of words in post}}, L=5$$

$$P(\text{post-A is cat-type}) = 1 - (1-p)^L = 1 - \left(1 - \frac{1}{5}\right)^5 = 1 - \left(\frac{4}{5}\right)^5 = 0.672$$

- For post-A; A: post contain word 'cute'
B: post is cat-type.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = P(A) = \frac{1}{5} \quad \left\{ \begin{array}{l} \text{Since, event 'A' and 'B' are independent} \\ P(A \cap B) = P(A) \cdot P(B) \end{array} \right.$$

$$P(B|A) = P(B) = 0.672$$

For post-C; A: post contain word 'cute'
B: post is cat type.

$$P(A|B) = P(A) = 0$$

$$P(B|A) = P(B) = 1 - (1-p)^L = 1 - \left(1 - \frac{1}{5}\right)^6 = 0.66$$

3. Designing a simple model for optimising upvotes :-

- $U(L) = \frac{-L^2}{20} + 3L$. {L = length of post, U = no. of upvotes}

$$U'(L) = \frac{-L}{10} + 3$$

$$U''(L) = \frac{-1}{10}$$

$$\text{for } U'(L) = 0 \Rightarrow L = 30$$

$$\text{at } L = 30 \rightarrow U''(L) = U''(30) < 0 \rightarrow \text{maxima.}$$

Thus, optimum length of posts for getting maximum upvotes $\Rightarrow L = 30$.

- $U(L) = \frac{-L^2}{20} + 3L$; $P(L, p) = 1 - (1-p)^L$

$$G(L, p) = P(L, p) \cdot U(L)$$

$$= \left(\frac{-L^2}{20} + 3L\right) \cdot [1 - (1-p)^L]$$